No evidence supports vitamin E indiscriminate supplementation.

For many years, the prevailing concept was that LDL oxidation plays the central role in atherogenesis. As a consequence, supplementation of antioxidants, particularly vitamin E, became very popular. Unfortunately, major randomized clinical trials yielded disappointing results and recent meta-analyses concluded that indiscriminate, high dose vitamin E supplementation results in increased mortality. This conclusion raised (quite reasonable) criticism, much of which referred to the characteristics of meta-analysis. In our recent study, we used a Markov-model approach, which is free of most of the limitations of meta-analyses. Our major finding was that the average quality-adjusted life years (QALY) of vitamin E- supplemented individuals was 0.30 QALY (95%CI 0.21 to 0.39) less than that of untreated people. In our view, this supports the view that indiscriminate supplementation of high dose vitamin E can not be recommended to the general public.In the present communication we address several recent studies that demonstrated negative effects of vitamin E and raise possible mechanisms that may be responsible for the harmful effects of vitamin E supplementation. We also review recent studies conducted with specific groups of patients that gained from vitamin E supplementation, indicating that although, on the average, indiscriminate supplementation of high dose vitamin E is not beneficial, specific populations may gain from vitamin E. The challenge is to establish selection criteria that will predict who is likely to benefit from vitamin E supplementation. Such criteria may be based either on the assumption that antioxidants are likely to be beneficial for people under oxidative stress or on knowledge regarding the benefit of sick people with certain diseases. In short, we adopt the view that vitamin E is a "double-edge sword" that should not be consumed until criteria are defined to predict who is likely to benefit from high dose supplementation of vitamin E.